Project Design Phase-II: Technology Stack (Architecture & Stack)

Date	28 October 2026
Team ID	N/A (Individual Project - Shriyanshu Jain)
Project Name	Global Electric Vehicle (EV) Market Performance & Insights
Maximum Marks	4 Marks

Technology Stack & Architecture Details:

S.No	Component	Description	Technology	Justification/Purpose
1	Data Source / Storage	Raw dataset containing global EV market records (sales, battery type, price, range).	CSV (Comma Separated Values)	Lightweight, universally supported format. Natively and quickly connects to Tableau Desktop/Public.
2	Data Pre-Processing	Handling missing values, data type formatting, and structuring train/test datasets.	Python (Pandas library)	Highly efficient for cleaning and combining large datasets (e.g., 100,000+ rows) before visualization.
3	Business Intelligence & Visualization	Creation of the 9 interactive visuals (scatter plots, heatmaps, bar charts) and dashboards.	Tableau Public / Tableau Desktop	Industry standard for interactive data visualization, supporting advanced calculations, trend lines, and storytelling.
4	Geographic Mapping Engine	Rendering filled maps to display EV manufacturing hubs and distribution.	Tableau Native Geocoding	Automatically interprets geographic fields (Country/Region) without needing external mapping APIs.
5	Presentation & Reporting	Delivery of the 6-8 slide executive briefing with voice-over instructions.	Microsoft PowerPoint	Standard corporate tool for summarizing personas (Emily, David, Sarah) and strategic recommendations.
6	Front-End /	Interface	Tableau	Provides a guided,

	User Interface	through which stakeholders interact with the data insights and visuals.	Storyboard	multi-point presentation flow that allows interactive filtering while maintaining a clear narrative.